

Intermediate Microeconomics — Week 2, Day 4

In-class Exercises: Monopoly and Price Discrimination

Perloff, Chs. 11–12 • work in pairs • solutions at the end

Exercise 1 — Single price, then perfect discrimination [competency: monopoly welfare]

A monopolist faces demand $P = 100 - 2Q$ and has constant marginal cost $MC = 20$ (no fixed cost).

- (a) *Single price.* Find the profit-maximising quantity and price, the firm's profit, consumer surplus, and the deadweight loss relative to the competitive ($P = MC$) outcome.
 - (b) *First-degree discrimination.* Now the monopolist charges each buyer their willingness to pay. Find the quantity produced, the firm's profit, consumer surplus, and deadweight loss. Contrast with part (a) in one sentence.
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Exercise 2 — Third-degree discrimination[competency: segmented pricing & elasticity]

A monopolist with constant $MC = 20$ sells in two separable markets:

$$\text{Market 1: } P_1 = 80 - 2Q_1, \quad \text{Market 2: } P_2 = 60 - Q_2.$$

- (a) Find the profit-maximising price and quantity in each market.
 - (b) Compute the price elasticity of demand at each optimum. Which market is charged the higher price, and is it the more or less elastic one?
 - (c) Verify the inverse-elasticity rule $\frac{P_i - MC}{P_i} = -\frac{1}{\varepsilon_i}$ in both markets.
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Reconvene: for Exercise 1 state what first-degree discrimination does to *efficiency* versus *who gets the surplus*; for Exercise 2, state the one-line pricing rule.